

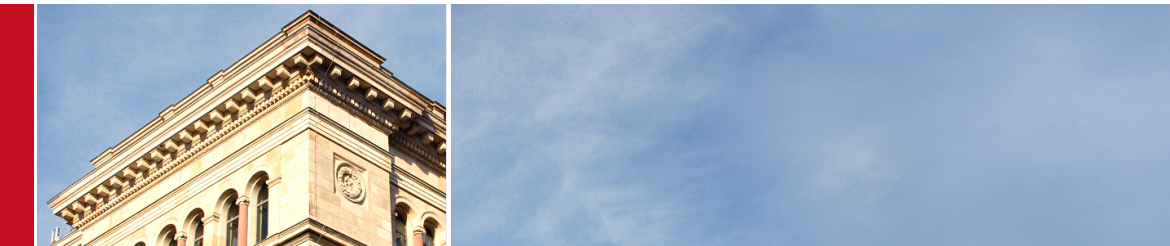
Beinert, R. and Bresch, J. (2024) – Denoising sphere-valued data by relaxed total variation regularization, **Proc. Appl. Math. Mech.**

Beinert, R. and Bresch, J. (2025) – Denoising hyperbolic-valued data by relaxed regularizations, **Lect. Notes Comput. Sci.**

Beinert, R., Bresch, J., and Steidl, G. (2025) – Denoising of sphere- and $SO(3)$ -valued data by relaxed Tikhonov regularization, **Inverse Probl. Imaging** **Berlin**

Beinert, R. and Bresch, J. (2026) – Denoising multi-color QR codes and Stiefel-valued data by relaxed regularizations, **Proc. Appl. Math. Mech.**

Bresch, J. (2026) – Denoising manifold-valued data by relaxed regularizations, **Oberwolfach Report No. 56/2025**



Denoising Manifold-Valued Data by Relaxed Regularizations

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Robert Beinert and Gabriele Steidl

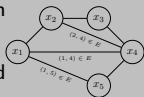
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Problem: Variational Inverse Problem

- **Denoising** signals on graph $G = (V, E)$



- Data lie on manifold $\mathbb{M} \subset \mathbb{R}^{d_{\mathbb{M}}}$

- Variational model with denoiser $\mathcal{D}$
 - a) Tikhonov regularization
 - b) Total Variation (TV)

$$\mathcal{D}(\cdot^1, \cdot^2) \in \{\| \cdot^1 - \cdot^2 \|_2^2, \| \cdot^1 - \cdot^2 \|_1\}$$

- Original problem:

$$\arg \min_{x \in \mathbb{M}^N} \frac{1}{2} \sum_{n \in V} \|x_n - y_n\|^2 + \lambda \sum_{(n,m) \in E} \mathcal{D}(x_n, x_m)$$

Second Idea: Convex Relaxation

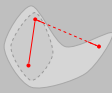
- **Neglect** $\text{rk}(Q_{(n,m)}), \text{rk}(V_n)$
- Relaxed **convex** problem with
 - a) Linear objective (Tikhonov)

$$\mathcal{D}(x_n, x_m) = \frac{1}{2} (x_n + x_m) - \ell_{(n,m)}$$

- b) Linear fidelity term (TV)



Main Difficulty: Nonconvexity



- Constraints $x_n \in \mathbb{M}$
- Lead to **nonconvex optimization problem**
- Hard to solve globally
- Standard convex methods cannot be applied directly

Third Idea: Simple Computations

- ADMM, due to (partial) linear
- critical points or prox_{TV} dim.-wise
- + projection on convex cone of PSD matrices → truncated SVD

General Framework, for many manifolds

- spheres $\mathbb{S}_{d-1}$
- rotation $\text{SO}(d)$
- hyperboloid $\mathbb{H}_d$
- binary $\mathbb{S}_0^d$ + theoretical “tightness”
- torus $\mathbb{T}_d$
- frames $\text{SE}(d)$
- Stiefel $\mathbb{V}_d(k)$

“tightness” can be observed in praxis

First Idea: (partial) Linearization

- Introduce auxiliary variables
- $\ell_{(n,m)} = \langle x_n, x_m \rangle, v_n = \|x_n\|^2$ and from def. of $\mathbb{M}$
- Encode constraints by **matrix representation** $Q_{(n,m)}, V_n \succeq 0$
- + minimal rank property $Q_{(n,m)}, V_n$
- Equivalent problem:

$$\arg \min_{x, v, \ell, \bullet} \sum_{n \in V} \frac{1}{2} v_n - \langle x_n, y_n \rangle + \lambda \sum_{(n,m) \in E} \mathcal{D}(x_n, x_m)$$

